

Roshni Kobula Raja

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SUMMARY

MS Computer Science graduate with hands-on software development experience in Python, C++, and JavaScript/TypeScript across backend services, full-stack applications, and generative AI systems. Comfortable designing retrieval and ranking algorithms, diagnosing issues to root cause, and writing validity, verification, and end-to-end tests.

EDUCATION

State University of New York at Binghamton, Master of Science (MS), Computer Science May 2026
Rajalakshmi Engineering College, B.Tech, Information Technology May 2024

EXPERIENCE

Graduate Assistant, Binghamton University, SUNY Aug 2025 - May 2026

- Architected an agentic AI system for career planning and role matching, focused on skill-gap analysis and personalized learning roadmaps with prompt engineering for role-fit explanations
- Partnered with Watson Career Services to gather requirements, test workflows, and iterate on agent behavior based on stakeholder feedback ahead of planned release for student use

Data Analyst Intern, Kanini Jun 2023 - Dec 2023

- Shipped a document-analysis application (Django, Flask) with PyTorch NLP pipelines, cutting manual processing time 40%
- Exposed REST APIs across the application for modular, reusable deployment across multiple workflows

AI/ML Intern, University of Texas at Dallas Jun 2022 - Aug 2022

- Trained a music-genre classification system using K-Nearest Neighbors and scikit-learn, achieving 82% accuracy
- Implemented the audio feature-extraction and model-evaluation pipeline in NumPy, testing and validating results

PROJECTS

EvidPath, AI Career-Planning Platform (Python, TypeScript, AWS) Mar 2026 - Present

- Created a full-stack career-planning platform with Python/FastAPI and GraphQL services on AWS EC2 and a React/TypeScript frontend, using RAG-based resume analysis and skill-gap scoring across 205 role pathways
- Wrote a 17-case pytest suite across 10 modules for validity and verification testing of retrieval quality, scoring accuracy, auth persistence, and security validation, plus Jest frontend tests and a smoke test against Docker Compose

CloudVault Lite, Deduplication Storage Engine (C++) May 2026 - Jul 2026

- Constructed a C++ content-addressable storage engine that uses SHA-256 hashing to detect and eliminate duplicate files, preventing redundant physical writes and improving storage efficiency
- Debugged a TOCTOU race condition in SQLite transactions to root cause and scaled concurrent uploads to 4 workers

Successor, Voice-Based Knowledge Capture for Semiconductor Fab Operations Jul 2026 - Aug 2026

- Designed a hybrid ranking algorithm on MongoDB Atlas that fuses Atlas Search and Vector Search results with Reciprocal Rank Fusion, surfacing fixes from spoken symptoms across 60 fault classes and 185 fix outcomes
- Orchestrated a multi-modal voice agent as a LangGraph state machine with ElevenLabs transcription and a MongoDB checkpoint, supporting stateful multi-step sessions and mid-session resumption

CivicTrace SF, Multi-Modal Video to Knowledge Graph Pipeline (Generative AI, Neo4j) Jul 2026

- Turned unstructured government meeting video into a structured Neo4j knowledge graph using multi-modal OpenAI and TwelveLabs models, flagging contradicted claims and approved projects with no permit on file
- Authored a Playwright end-to-end suite with Chromium and mobile projects, retries, and trace-on-failure, plus pytest coverage for backend routes, containerized with Docker Compose

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, TypeScript, SQL

Software Engineering: Data structures and algorithms, REST and GraphQL APIs, FastAPI, Django, React, root cause analysis

Testing & Infra: pytest, Playwright, Jest, smoke and end-to-end testing, Git, Docker, Kubernetes, AWS EC2, Linux

Generative AI & Data: LLMs, multi-modal models, RAG, LangGraph, MCP, PyTorch, PostgreSQL, MongoDB, ClickHouse, Neo4j

ACHIEVEMENTS & CERTIFICATIONS

- Best Product and Best Use of Aria, CoreWeave Hackathon, Sep 2026
- 2nd Place, Better Days Hackathon (ClickHouse x LibreChat) for Silent Flood, Aug 2026
- Oracle Agentic AI Foundation Certificate, Oracle, Aug 2026
- Women in Tech Salesforce Training Program, MTX, Nov 2025
- Robotic Process Automation, UiPath, Jul 2023